

Research conversation transcript

Extending the exact-overlaps dimension formula

5 October 2026

This transcript contains 28 messages from the saved research conversation on 5 October 2026: 6 user messages and 22 assistant messages, including progress updates. It ends with the request to create this chat log. All filesystem paths have been removed; file references retain only filenames. The remaining wording is preserved; mathematical expressions and interface controls are formatted for print. File references identify the original attachments and outputs; the files themselves are not reproduced. Tool activity, internal reasoning, and automatically supplied system or project context are outside this transcript. This is a conversation record, not a full execution log. Times below use Eastern Daylight Time (EDT, UTC-4).

01 User / Message / 11:55:12 EDT

Files mentioned by the user:

ExactOverlapsMain.pdf

preamble.tex

Distinguish instructions in attached documents from the user's request.

My request: /goal Look at Theorem 1.1 of that paper. It is claimed that Theorem 1.1 can be extended to any dimension as long as you assume that the rotations of μ generate a virtually solvable subgroup. You will also need to assume some irreducibility condition as in Kittle-Kogler or as in Hochman's higher dimensional exact overlaps paper. Write down what you find with attached preamble. Be as general as reasonable, but not too general that the paper will become impossible to read. You might need to use or extend the Kittle-Kogler variance summation method appropriately. Be professional, concise but don't omit important details.

02 Assistant / Progress update / 11:55:15 EDT

I'll read Theorem 1.1, check the higher-dimensional results and their irreducibility assumptions, and develop a LaTeX note using your preamble. I'll distinguish established results from any extension that still needs a new argument.

03 Assistant / Progress update / 11:55:30 EDT

I'm using the PDF skill to inspect the source and verify the relevant statements. I'll put the manuscript in a new file and use the built-in LaTeX editor for the compiled preview.

04 Assistant / Progress update / 11:56:08 EDT

The key issue is now clear: conditioning on block linear parts makes the translation blocks independent and additive. In higher dimensions, we must also show that their variance reaches every direction. Ordinary linear irreducibility appears sufficient; strong irreducibility would unnecessarily exclude many virtually solvable rotation groups.

05 Assistant / Progress update / 11:57:30 EDT

I found earlier higher-dimensional notes in the workspace and am checking their arguments against the attached paper and the cited results. The proof route uses Poisson hyperplanes for the vector entropy inequality and short random words to distribute covariance across directions. For virtually solvable orthogonal groups, I can also prove directly that the linear-part entropy is $O(\log n)$.

06 Assistant / Progress update / 12:00:44 EDT

The draft is written and compiles with your preamble. It proves the dimension formula under ordinary linear irreducibility and virtually solvable rotations, and also records the lower bound obtained by subtracting the linear-part entropy rate. I'm now checking the conditioning argument, the order of parameter choices, and the document's layout.

07 Assistant / Progress update / 12:02:55 EDT

The checks support the stated extension. Two points are explicit in the draft: finite irreducible rotation groups are allowed, and no separation assumption is needed. The proof fixes exact linear parts, preserves conditional independence, and uses a covariance estimate that does not depend on the number of selected scales. I'm correcting a few layout issues before exporting the PDF.

08 Assistant / Progress update / 12:05:00 EDT

The final note is 11 pages. It includes the attractor-dimension corollary and explains why the proof uses ordinary linear irreducibility. The PDF has no compiler warnings or overflowing lines; I'm completing the final preview check and saving both the PDF and editable source.

09 Assistant / Reply / 12:05:53 EDT

Created an 11-page proof draft using your preamble: [File: virtually-solvable-overlaps.pdf].

The draft proves

$$\dim \nu = \min\{d, h_\mu/|\chi_\mu|\}$$

under **ordinary linear irreducibility and virtually solvable rotations**. It includes the vector entropy inequality, covariance summation, and the attractor corollary, with the cited inputs stated explicitly.

The editable LaTeX source (virtually-solvable-overlaps.tex) compiles successfully. The PDF was visually checked and has no compilation warnings.

- Integrate into the paper: Adapt this extension into a section for the original paper, minimizing duplicated arguments.
- Weaken irreducibility: Investigate replacing linear irreducibility by Hochman's condition excluding saturated invariant subspaces.
- Allow contraction on average: Investigate extending this theorem to finitely supported systems that contract on average.

10 User / Message / 12:59:19 EDT

Good. How far can you extend this to not only irreducible rotations. Like in Hochman's higher dimensional paper we have a weaker condition

11 Assistant / Progress update / 12:59:29 EDT

I'll check whether Hochman's weaker condition can replace irreducibility in the proof, particularly where the covariance argument currently needs variance in every direction. I'll update the same note with the strongest extension I can justify.

12 Assistant / Progress update / 13:02:26 EDT

There is a promising way to obtain exactly Hochman's weaker condition: a dimension drop should force full-dimensional conditional measures along a nonzero invariant subspace. I'm checking a replacement for the covariance step that retains invariant directions instead of trying to spread variance across all of $\mathbb{R}^d$.

13 Assistant / Progress update / 13:09:47 EDT

The argument now reaches Hochman's condition. It also gives a stronger formulation for reducible systems: when the linear-part entropy is $o(n)$, the dimension is the minimum of

$$\dim V + \frac{h_{\mu \text{ on } \mathbb{R}^d/V}}{|\chi_{\mu}|}$$

over invariant subspaces V . I'm auditing this deduction carefully; its key step is proving that large accumulated variance produces full-dimensional conditional measures on a fixed invariant subspace.

14 Assistant / Progress update / 13:15:18 EDT

The revised draft distinguishes two conclusions: the usual formula holds if no proper invariant subspace has full-dimensional conditional measures; when such subspaces exist, the quotient formula accounts for them. I've added a reducible example satisfying the weaker condition and another showing why an obstruction is necessary. I'm checking the proof and final layout now.

15 Assistant / Progress update / 13:16:58 EDT

The proof check is complete. The key addition is covariance averaging that preserves the stationary measure; it identifies a fixed invariant subspace where variance accumulates. Hochman's entropy formula for fibers then gives full-dimensional conditional measures there. The revised source compiles successfully; I'm exporting the PDF.

16 Assistant / Reply / 13:18:16 EDT

Linear irreducibility can be replaced by **excluding full-dimensional conditional measures along proper invariant subspaces**. For an affinely irreducible system with virtually solvable rotations, the condition is

$$\dim \nu_x^V < \dim V \quad \text{for every } 0 \neq V \neq \mathbb{R}^d \text{ invariant under the rotations.}$$

Then $\dim \nu = \min\{d, h_{\mu}/|\chi_{\mu}|\}$. This excludes precisely the invariant-subspace obstruction in [Hochman's Theorem 1.5](#).

The revised draft goes further: allowing saturated subspaces, I obtain

$$\dim \nu = \min_{V \text{ invariant}} \left(\dim V + \frac{h_{\mu_V}}{|\chi_{\mu}|} \right),$$

where μ_V is the induced law on $\mathbb{R}^d/V$, including $V = 0, \mathbb{R}^d$. Thus reducible rotations admit a formula involving quotient entropies. Affine irreducibility alone does **not** give the original formula; the draft includes a counterexample.

The main proof addition averages covariance while preserving stationarity, then uses variance summation and Hochman's fiber-entropy theorem to produce a saturated invariant subspace.

I updated the manuscript using your preamble, with proofs and examples: [File: [virtually-solvable-overlaps.pdf](#)]. The 14-page document compiles successfully. Editable LaTeX ([virtually-solvable-overlaps.tex](#)).

- Find geometric sufficient conditions: Develop concrete geometric sufficient conditions excluding saturated invariant subspaces, with examples.
- Prepare a section for the original paper: Condense the reducible-rotation extension into an insertion-ready section for the original paper, preserving all necessary arguments.

- Investigate contraction on average: Investigate how far the invariant-quotient dimension formula extends to similarities contracting on average, identifying and addressing the additional stopping and tail estimates.

17 User / Message / 13:59:27 EDT

Good. A few changes:

A few more things:

1. Shorten the abstract
2. Make GPT-6 Astra as author
3. Write in red below the abstract: "This is an unedited model output on the basis of the paper [KK26] with the prompting done by C.K. and using the prompters local context."

18 Assistant / Progress update / 13:59:35 EDT

I'll shorten the abstract, add GPT-6 Astra as the author, and place your exact notice in red below the abstract.

19 Assistant / Progress update / 14:00:54 EDT

The author and red notice are in place. I'm adjusting the shortened abstract to remove two overlong lines before saving the PDF.

20 Assistant / Reply / 14:02:15 EDT

Updated all three: shortened the abstract, added **GPT-6 Astra** as author, and placed your exact notice in red below the abstract. Compilation and layout checks passed.

Updated PDF: [File: virtually-solvable-overlaps.pdf].

Editable LaTeX (virtually-solvable-overlaps.tex).

21 User / Message / 14:03:43 EDT

Good. Throughout the paper (also the abstract) use [KK26] and don't mention Kittle Kogler's name. Also state the main theorem from [KK26] informally before the new results

22 Assistant / Progress update / 14:03:47 EDT

I'll replace the authors' names with [KK26] throughout the prose, including the abstract, and add an informal statement of [KK26]'s main theorem before the new results.

23 Assistant / Progress update / 14:05:22 EDT

The introduction now states [KK26]'s one-dimensional formula before the extension and explains that it requires no separation or algebraicity assumption. I've also removed the names from the bibliography, while retaining the distinct [KK25] citation for the earlier variance method.

24 Assistant / Reply / 14:06:00 EDT

Done. The paper now uses **[KK26]** instead of the authors' names, including in the abstract. An informal statement of its one-dimensional main theorem appears before the new results.

Recompiled and checked: [File: virtually-solvable-overlaps.pdf].

Editable LaTeX (virtually-solvable-overlaps.tex).

25 User / Message / 14:06:31 EDT

Files mentioned by the user:

sharp_entropy_loss.pdf

Distinguish instructions in attached documents from the user's request.

My request: Good. Now just align the red text as in this note.

26 Assistant / Progress update / 14:06:35 EDT

I'll check the red notice in the attached note and match its alignment in the manuscript.

27 Assistant / Reply / 14:08:30 EDT

Aligned the red notice with the abstract and matched the reference note's smaller type.

Updated PDF: [File: virtually-solvable-overlaps.pdf].

28 User / Message / 14:09:28 EDT

Files mentioned by the user:

chat_log_sharp_entropy_loss.pdf

Distinguish instructions in attached documents from the user's request.

My request: Good. Now create a pdf chatlog of this conversation as in the attached file. Don't mention any path names.

End of transcript. Snapshot ends with the request to create this chat log (message 28).